

# Exploring the Applications of IoT

The Internet of Things has become a game changer in today's world and is playing a vital role in the industrial revolution. This article explores the uses of IoT in healthcare systems, water supply, recycling systems and many other areas.

**T**he Internet of Things (IoT) could be a game changer for India, but the country is lagging far behind China in its adoption. Today, the IoT is transforming industrial revolution. An IIoT factory can yield 5-12.5% cost reduction in operation and optimisation, 10-40% cost savings in predictive maintenance and 20-50% cost reduction in inventory optimisation.

To boost IoT concepts, Indian technology giant Wipro announced the launch of industrial IoT Center Of Excellence (COE) in Kochi, Kerala on March 14, 2019. According to Wipro,

this COE aims at developing innovative solutions for customers in industrial manufacturing, automotive, healthcare and pharmacy. It aims at developing customer-ready IoT solutions.

## Innovations needed for IoT industries

Innovation leads, the world follows. With the rise in the Indian Startup Ecosystem, innovation has reached its peak in the country with 45,444 patent applications filled in 2016-17 alone.

Today, a lot of IoT related startups are coming and working in India. They are taking advantage of the Internet to enhance the abilities of the devices. The devices may be lighting system, entry into home, water sprinklers used for irrigation in agriculture, traffic control system, road lighting systems and many more. In a smart city, electric transmission, billing and the security of various services such as banks, ATMs, hotels, etc, are all done through IoT-operated devices. Dealing with IoT related projects by companies and startups is offering great challenges.

Vyuti is a visual object intelligence provider for the manipulation of objects by robotic arms. It can pick, orient and place objects in various locations. With adequate training, the solutions can be applied in sectors like manufacturing, construction, healthcare, and warehousing. Today, there are many solutions in the market for object recognition. Funded by N.A. Gokul, Vyuti is in talks with TVS, AMS and Sansera.

Numocity is working with the electric vehicle ecosystem. It aims to connect batteries, charging stations, etc, via a transaction engine, distributed ledger and an artificial intelligence (AI) based engine. Founded by Ravikiran Annasamy, Numocity was a member of Bosch's DNA 2.0 Cohort in 2018.

## Areas benefitting through IoT

While India is considered to be the fourth largest home for startups, the failure rates are also very high here due to many factors including difficulties in managing the ventures.

The hundred billion dollars Indian healthcare industry is set for revolution. Big Data, IoT and AI will shape the future of this new ecosystem.

Increased entry of affordable IoT devices and rising consumer expectations are driving the rapid growth of the IoT market. The IoT market in India is expected to reach US\$ 15 billion by 2020, accounting for nearly five per cent of the total global market. With nearly 120 firms offering solutions in the IoT segment today, there are tremendous opportunities for further growth.

Smart lifestyles, embedded homes, buildings, healthcare and manufacturing are key sectors in India for IoT startups. Investors are now showing more interest



Wipro's IIoT Center of Excellence in Kochi, Kerala (Credit: Tech Observer)

